KneeVision

Team S.T.A.R Labs

Agenda

01 SPRINT 0 RECAP

- Research Team
- Improvements
- Project Description
- Team Working Agreement
- Personas

02 PROJECT OVERVIEW

- MVP
- Languages and Tools
- Algorithms
- Diagrams

03 PROJECT BACKLOG

- User Stories
- Sprint 1 Stories
- Test Cases

04 TEAM METRICS

- Team Velocity
- Burndown Charts
- Completed/ Committed Ratio

05 PRODUCT

- Sprint 1 Retrospective
- Sprint 2 Planning
- API
- Live Application Demo

Our Research Team

Daniel Fox
Team Leader/ Developer

Manohar Killamsetti
Scrum Master/Developer

Siming Li
Developer/Tester

Sarvesh Shah
Cloud Developer

Swatej Goud
Backend Developer

Deep Patel
ML Developer

Improvements

- Project Description Page added (slide 7)
- Updated Tools & Technologies slide to include revised choices (slide 15)
- Added Technologies and Algorithms explanations (slides 16 – 20)

Problem Statement

Knee X-ray imaging is commonly used to evaluate osteoarthritis, yet interpreting radiographic signs of disease severity can be subjective and inconsistent due to subtle visual differences and reliance on manual assessment. While advanced imaging techniques such as MRI provide greater detail, they are expensive and less accessible. This creates a need for scalable and consistent analysis methods that can assist in assessing knee osteoarthritis using widely available X-ray images.

Project Definition

- A web-based platform that uses deep learning to analyze knee X-ray images for knee osteoarthritis severity assessment.
- Intended for individuals, researchers, and clinicians seeking consistent and scalable analysis of knee X-ray data.
- Leverages publicly available datasets and established convolutional neural network (CNN) models.
- Designed as an AI-assisted decision-support and research tool, not a diagnostic system.
- Benefits include improved consistency, scalability, and reduced subjectivity in manual X-ray interpretation.

Project Description

Project Name:	KneeVision
Team:	S.T.A.R. Labs
Project Description:	KneeVision is a web-based platform that uses machine learning to analyze knee X-ray images, enabling faster and more accessible diagnostic insights. **For** researchers and medical professionals **who** upload knee X-ray images **the** KneeVision app **is a** web-based application that uses machine learning to analyze knee X-rays and assist in detecting potential conditions **that** provides faster, data-driven insights to support diagnosis and decision-making **unlike** traditional workflows that rely solely on manual review by specialists and can be time-consuming **our application** improves accessibility and efficiency by enabling quick image analysis and centralized result management
Benefit Outcomes:	• Faster turnaround time for preliminary image analysis • Improved accessibility to diagnostic insights • Centralized storage of user-specific medical images • Enhanced workflow for reviewing and managing X-ray data • Scalable foundation for integrating advanced AI/ML models
Github Link:	https://github.com/htmw/STARLABS

Team Agreement: Collaboration Standards

Meeting Pattern

Synchronous sessions twice weekly (Sun/Tue).
Mandated Agenda
documents for every session.
Continuous async sync via WhatsApp/Discord.

Decision Making

Majority-vote consensus based on researched evidence. Inclusive voicing of concerns during all discussion phases.

Technical Workflow

No direct pushes to Main. Isolated feature branches only.
Peer Approval. Every PR requires at least one secondary reviewer.
Jira Hygiene. Mandatory linking of work to Jira tracker IDs.
Blockers. Resolve locally & document attempts before escalation.

Team Agreement: Definition of “Done”

Category	Standard for Completion
Execution	Code runs without syntactical or runtime errors.
Compliance	Work meets all Industry Standards.
Verification	Work reviewed by multiple members & verified on multiple systems.
Documentation	All necessary docs tagged or updated along with code push.
Testing	All pre-defined test scenarios must be passed.

Persona - 1: The Practitioner

ORTHOPEDIC SURGEON | URBAN TEACHING HOSPITAL

Dr. Sarah Chen, 42

An expert in orthopedic medicine with 15 years of experience. Sarah operates in high-volume clinics where diagnostic speed is paramount to patient throughput and surgical triage.

KEY GOALS

- Rapid knee X-ray triage
- Reduce initial assessment time
- Validation for borderline cases

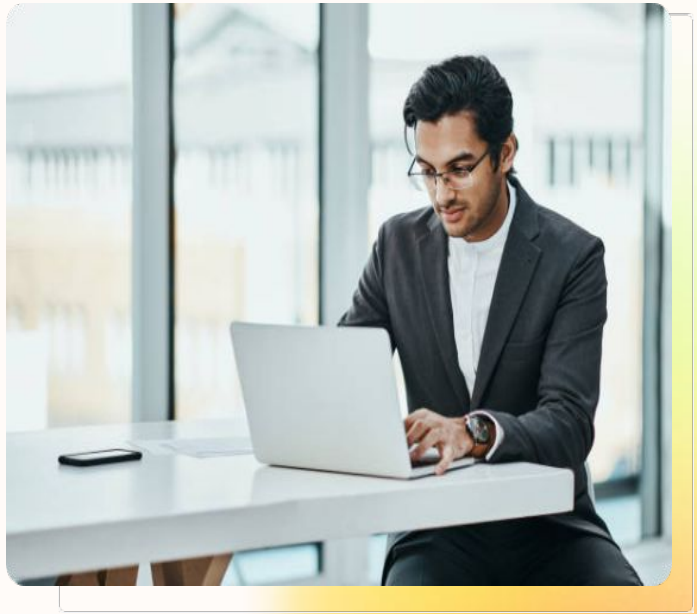
- Fatigue from patient volume
- Reader reporting variability
- Limited case analysis time

Persona - 2: The Research Coordinator

CLINICAL RESEARCH COORDINATOR

Alex Rodriguez, 29

A data-centric professional managing multi-site studies. Alex focuses on standardization, data integrity, and inter-rater reliability across large datasets.



STRATEGIC GOALS

- Standardize severity grading
- Efficient batch processing
- Structured data exporting
- Ensure rater reliability

OPERATIONAL PAINS

- Manual grading inconsistency
- Lengthy training cycles
- Audit trail requirements
- Historical data re-analysis

Persona - 3: The Learner

3RD YEAR MEDICAL STUDENT

Maya Patel, 24

A digital native in her clinical rotations. Maya uses the platform as a pedagogical tool to build pattern recognition skills and diagnostic confidence before her residency.

LEARNING GOALS

- Identify radiographic signs
- Immediate feedback practice
- Grade/Clinical correlation

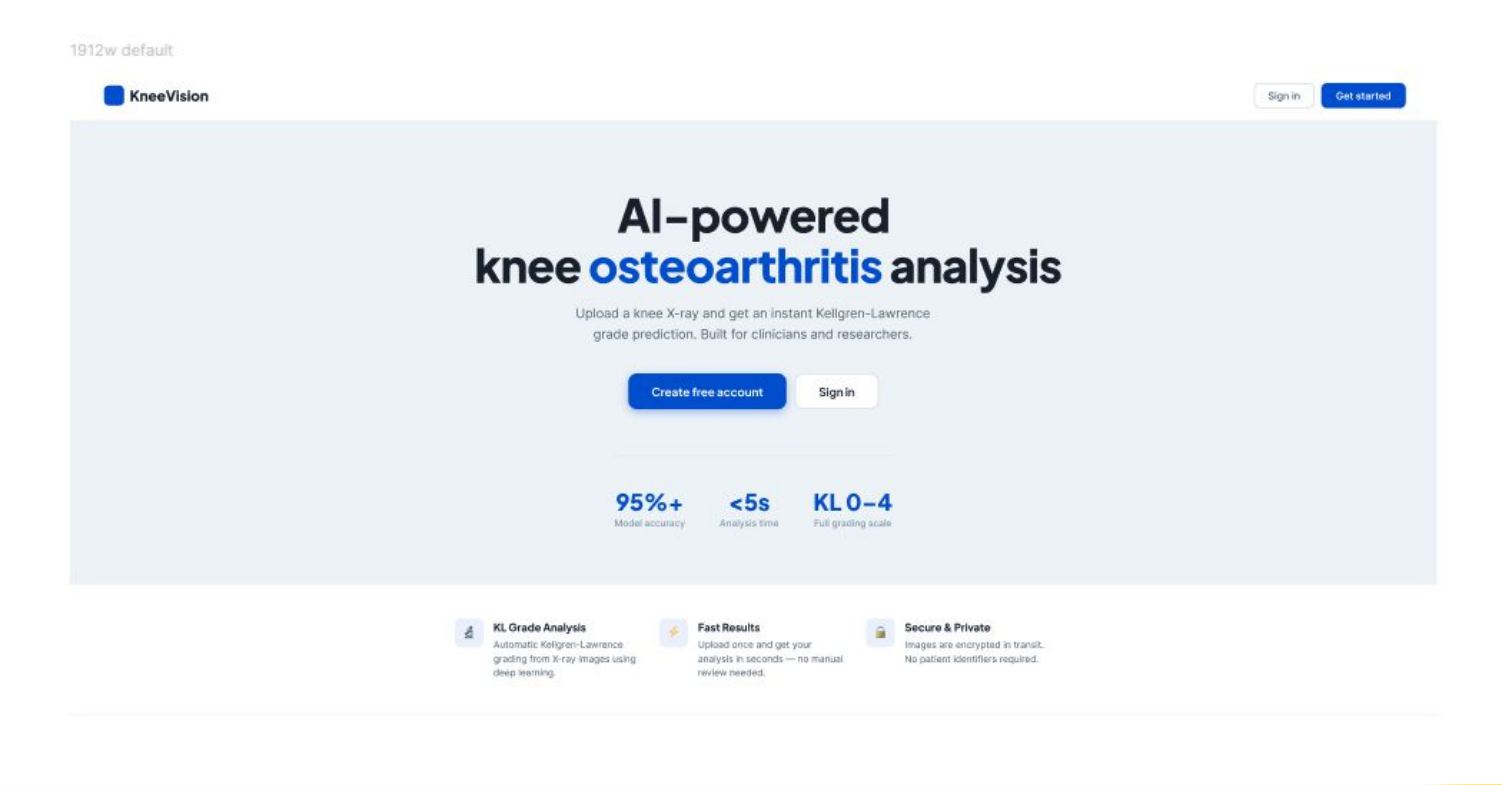
ACCESS OBSTACLES

- Limited diverse case access
- Assessment uncertainty
- Limited attending time

What will our MVP be?

- Web-based interface
- Image preprocessing pipeline
- Deep learning model
- Severity assessment output
- Basic result visualization
- Mobile-responsive UI

Project Design (Figma)



Tools & Technologies

Languages: 

Frontend: 

Backend: 

ML/DL Libraries: 

Database: 

Cloud: 

Tools: 

ML Inference: 

Misc. Libraries: 

FRONTEND

React · Vite · Tailwind CSS · TypeScript

React 18

Component-based UI

- Open-source JavaScript library for interactive UIs
- Reusable component architecture for maintainability
- State management with useState and useEffect hooks
- Pages: Landing, Login, Register, Upload, Gallery

Vite

Build toolchain

- Lightning-fast HMR and dev server startup
- Instant compilation during development
- Optimised production builds with tree-shaking
- Configured with Vite proxy to backend

Tailwind CSS

Utility-first styling

- Direct utility approach replaces custom CSS files
- Consistent design tokens across all components
- Responsive layouts with mobile-first breakpoints
- Dark-mode ready with zero extra config

TypeScript

Type-safe JavaScript

- Static typing catches errors at compile time
- Typed props and API response interfaces
- Improves IDE autocomplete and refactoring
- Compiled to JavaScript for browser execution

BACKEND

Node.js · Express.js · JWT · bcrypt

Node.js

Runtime environment

JavaScript runtime built on Chrome's V8 engine. Enables server-side JS execution with non-blocking I/O for high concurrency — perfect for handling multiple concurrent upload and analysis requests.

Express.js

Web framework

Minimal and flexible Node.js web framework. Handles all REST API routes: auth (register/login), image presign, raw file upload, image metadata registration, and image retrieval with JWT middleware.

JWT Auth

Stateless authentication

JSON Web Tokens secure every protected route via the requireAuth middleware. Tokens are signed with a secret, expire in 1 hour, and carry userId + email so any route knows who made the request.

bcrypt

Password hashing

Industry-standard adaptive hashing algorithm. Passwords are never stored in plain text — bcrypt hashes with salt rounds of 10, making brute-force attacks computationally infeasible.

DATABASE & CLOUD

MongoDB · AWS · Local filesystem

MongoDB

NoSQL document database

users collection

Stores email, hashed password, userId, timestamps

images collection

fileUrl, originalName, contentType, uploadedBy { userId, email }

predictions collection

imageId FK, klGrade, confidence, probabilities, modelVersion

Indexing

Unique index on users.email prevents duplicate accounts

AWS / Cloud

Cloud infrastructure (planned)

Amazon S3

Object storage for uploaded X-ray images at scale

EC2 / Lambda

Scalable compute for Express backend and ML inference

CloudFront

CDN for fast global delivery of static assets

Current dev

Local filesystem /uploads/ used during development

DEVOPS & TOOLS

Docker · GitHub · Jira · VS Code · Jest · Vitest

Docker

Containerises the Express backend and ML service so the environment is identical across dev, staging, and production machines.

GitHub

Version control and collaboration. Feature branches, pull requests, and code review. CI/CD pipeline triggers test suite on every push.

Jira

Agile project management. SCRUM board tracks sprints (SCRUM-18 register, SCRUM-19 login, SCRUM-20 image upload). Backlog prioritised by story points.

VS Code

Primary IDE with TypeScript, ESLint, and Prettier extensions. Integrated terminal runs both Vite dev server and Express backend simultaneously.

Jest + Vitest

Backend unit and integration tests with Jest (86 tests, ~89% coverage). Frontend component tests with Vitest + React Testing Library.

ESLint + Prettier

Code quality enforced at save. ESLint catches bugs, Prettier formats consistently. Husky pre-commit hooks block bad commits.

Model Pipeline Explanation

Dataset: Mendeley Dataset (8260 Images)

Model Plan:

Cleaning Dataset (Manually deleting the the defective images and cropping to ROI [Region of Interest])

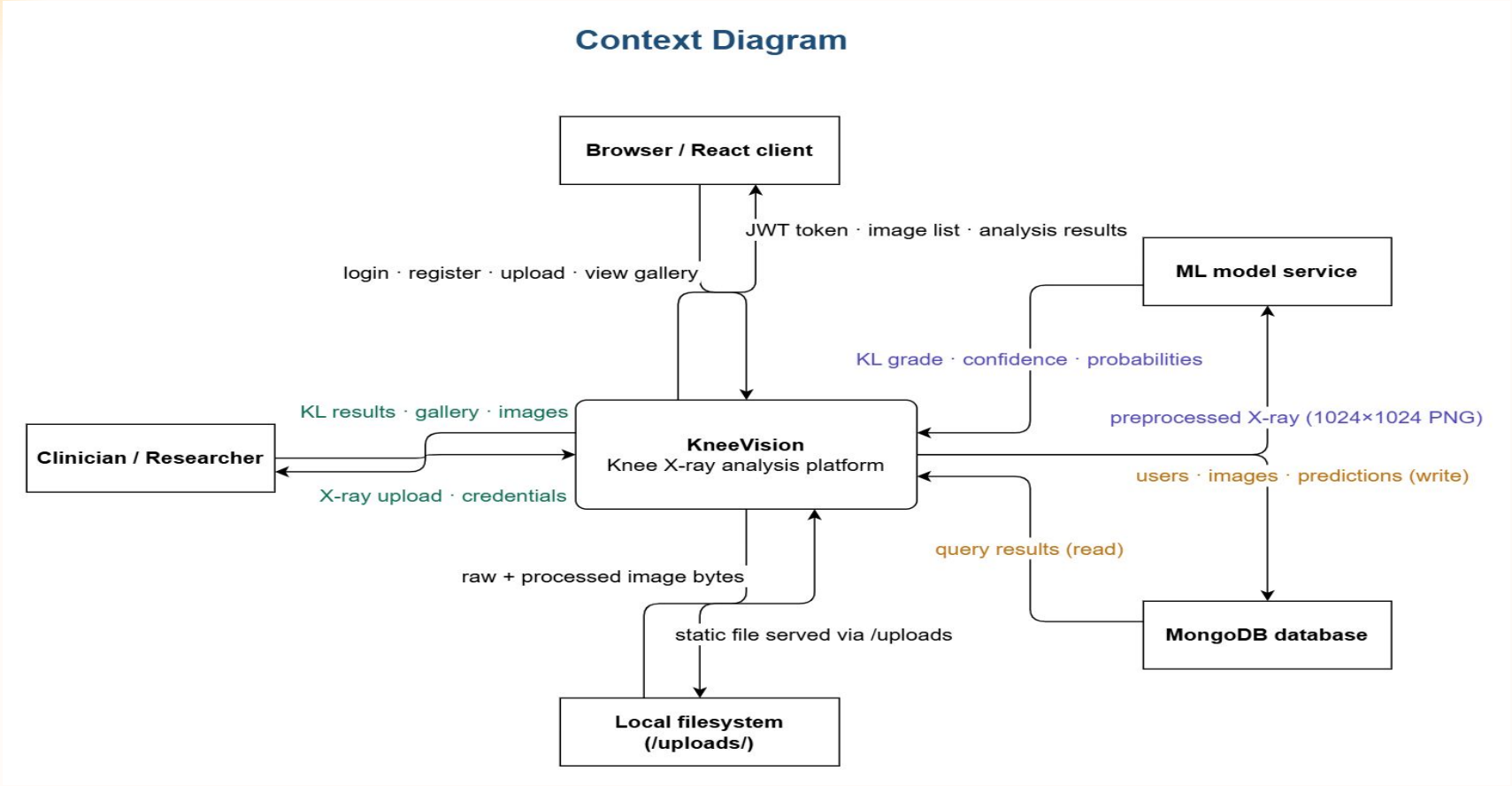
Generating synthetic data using DCGAN (Deep Convolutional Generative Adverserial Network)

Preprocessing (Enhancing images using Adaptive Histogram Equalization [AHE])

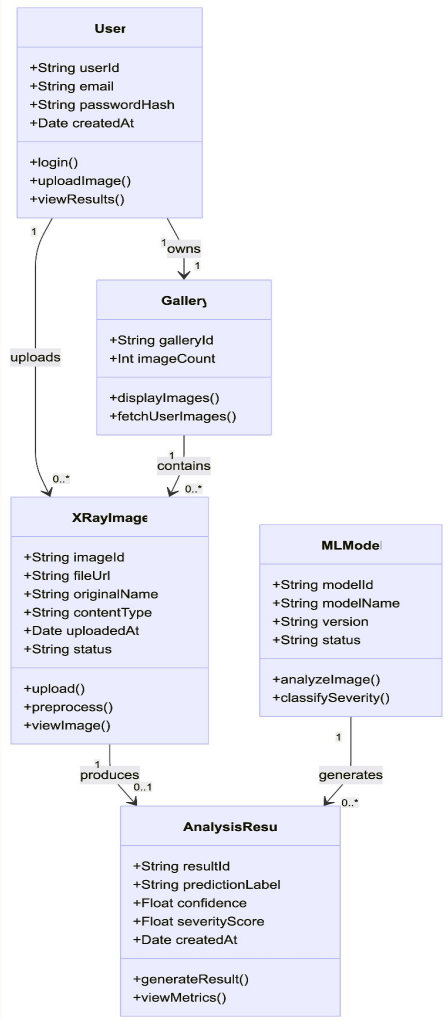
Using Pre-Trained Models (Example: VGG16, ResNet50, etc.)

Metrics Evaluation

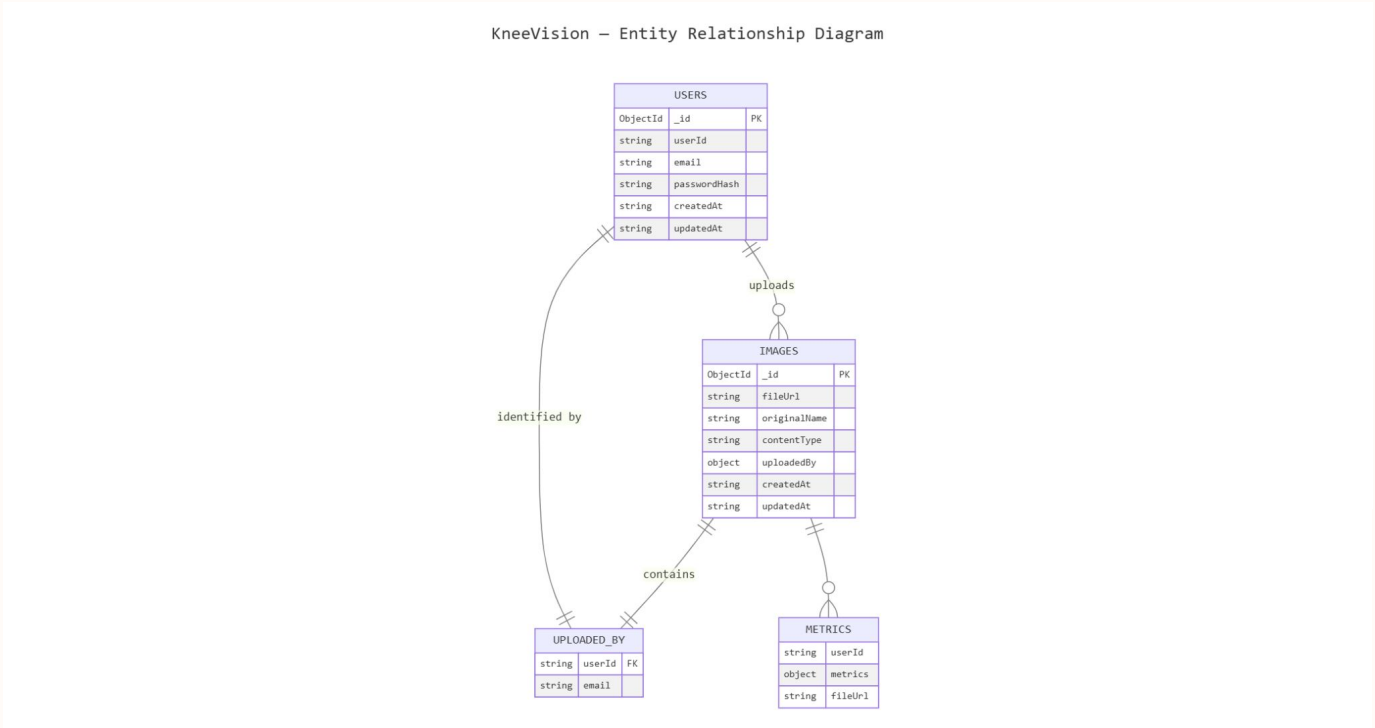
Context Diagram



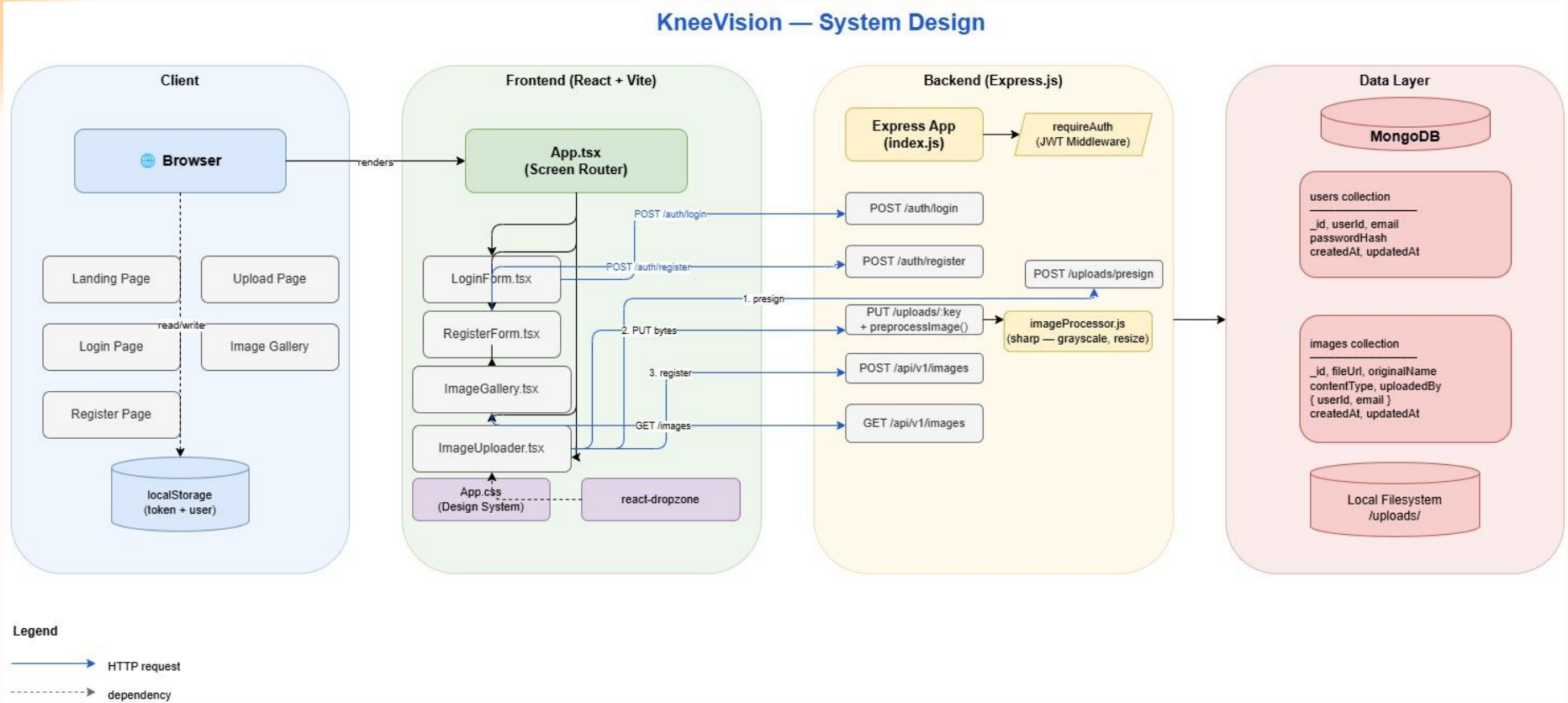
Class Diagram



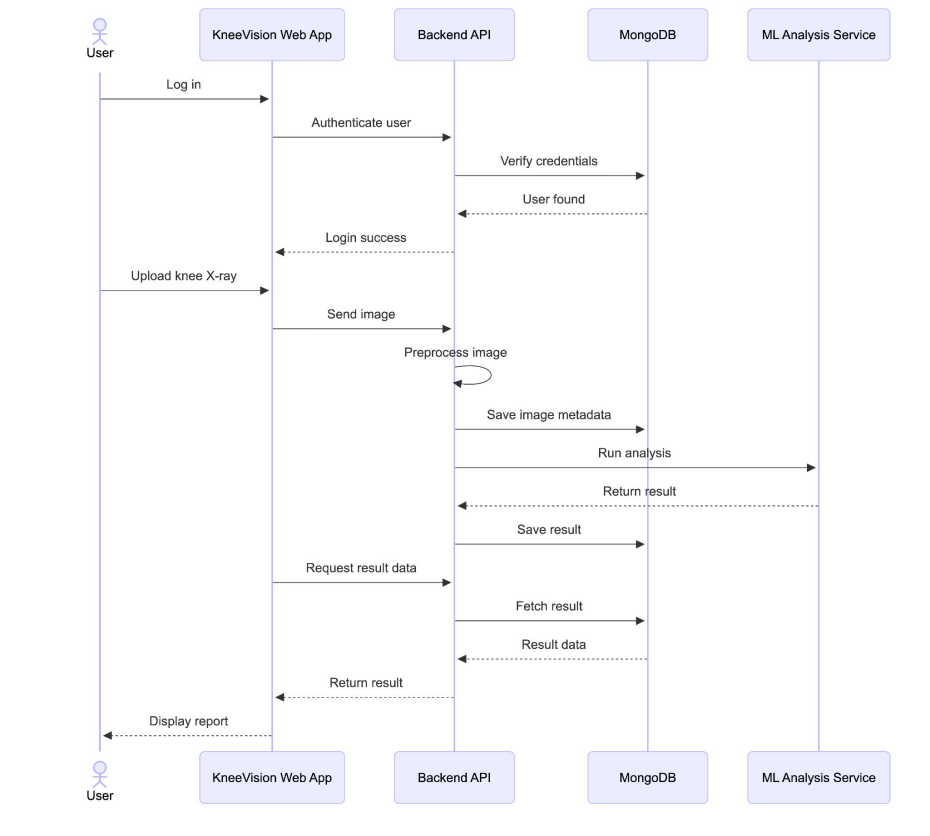
ER Diagram


KneeVision — Entity Relationship Diagram
USERS
ObjectId _id PK
string userId
string email
string passwordHash
string createdAt
string updatedAt
uploads
IMAGES
ObjectId _id PK
string fileUrl
string originalName
string contentType
object uploadedBy
string createdAt
string updatedAt
identified by
contains
UPLOADED_BY
string userId FK
string email
METRICS
string userId
object metrics
string fileUrl

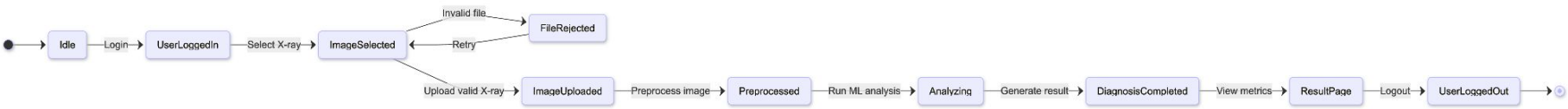
Architecture Diagram



Sequence Diagram



State Diagram



PROJECT BACKLOG - USER STORIES

US_ID	User Stories	Acceptance Criteria	Feature	Story Points
US_01	As an individual user, I want to upload knee X-ray images so that I can receive an AI-based assessment of osteoarthritis severity.	The patient can access an interface to upload an image. The patient can select and submit an image from device. The system accepts image files in specified formats: JPEG, JPG, PNG, DICOM. The uploaded image is stored and associated with the patient's case. A confirmation message is displayed upon successful upload.	Image Upload	**13**
US_02	As a research coordinator, I want to upload multiple X-ray images at once so that I can analyze datasets efficiently.	The patient can upload multiple images (up to 3) to one case. The system accepts multiple images in specified formats (JPEG, JPG, PNG). All uploaded images are associated with the same patient case. The patient receives a confirmation message after uploading each image. The system displays a preview of all uploaded images for review before analysis.	Image Upload	**8**

PROJECT BACKLOG - USER STORIES

US_ID	User Stories	Acceptance Criteria	Feature	Story Points
US_03	As an individual user, I want to register an account so that I can store my basic information and access personalised services.	The user can access a registration page requiring: Full name, Email address, Password creation. Post-registration, the user can log in and update their profile information.	User Management	**2**
US_04	As an individual user, I want to log in to my account so that I can manage my cases and view diagnoses.	The user accesses a login page requiring: Email address, Password. Upon successful login, the user is directed to their dashboard.	User Management	**1**
US_05	As a user, I want to view my past cases and track their statuses so that I can monitor my health progress.	The user can access a 'Case History' section after logging in. All past cases are listed in chronological order (most recent first). Each case displays: case ID, case open date, status. The system updates case statuses in real-time. Clicking on a case entry navigates the user to that case's report.	User Management	**2**
US_06	As an individual user, I want to implement JWT authentication so that my session is secure.	JWT tokens are generated on login and validated on every protected route. Tokens expire after a defined period and require re-authentication. Invalid or expired tokens return a 401 Unauthorized response.	User Management	**3**

PROJECT BACKLOG - USER STORIES

US_ID	User Stories	Acceptance Criteria	Feature	Story Points
US_07	As an individual user, I want to receive an OA severity diagnosis so that I know the knee condition I am facing.	After submitting the image, the user receives a diagnosis. The diagnosis is presented as the KL grade (0–4) and severity label. The system provides the diagnosis within a reasonable time (< 1 min). The diagnosis is stored and associated with the user's case.	AI Model Integration	20
US_08	As an individual user, I want to receive the most accurate possible diagnosis so that I can trust the assessment provided.	The system continuously improves the model and algorithms for better diagnostic results. The diagnosis accuracy is optimised based on model refinements and external validation. The confidence level of the diagnosis should be high (e.g., 95%).	AI Model Integration	11
US_09	As an individual user, I want to receive a brief description of the diagnosed condition so that I can understand the issue I am facing.	After receiving the diagnosis, the user is provided with a brief description of the diagnosed condition. The description is written in clear, understandable language. The description includes common symptoms and causes.	AI Model Integration	2
US_10	As an individual user, I want to receive treatment recommendations so that I can take action towards treating my knee condition.	After receiving the diagnosis, the user is provided with treatment recommendations. The description is written in clear, understandable language. The description includes common symptoms and causes of the condition.	AI Model Integration	3

Sprint 1 - USER STORIES

US_ID	User Stories	Acceptance Criteria	Feature
US_01	As an individual user, I want to upload knee X-ray images so that I can receive an AI-based assessment of osteoarthritis severity.	The patient can access an interface to upload an image. The patient can select and submit an image from device. The system accepts image files in specified formats: JPEG, JPG, PNG, DICOM. The uploaded image is stored and associated with the patient's case. A confirmation message is displayed upon successful upload.	Image Upload
US_07	As an individual user, I want to receive an OA severity diagnosis so that I know the knee condition I am facing.	After submitting the image, the user receives a diagnosis. The diagnosis is presented as the KL grade (0–4) and severity label. The system provides the diagnosis within a reasonable time (< 1 min). The diagnosis is stored and associated with the user's case.	AI Model Integration

SPRINT 1 STORIES COMPLETED AND NOT

US_ID	User Stories	Completed Date	Story Points	Completed
US_01	**As an individual user, I want to upload knee X-ray images so that I can receive an AI-based assessment.**		31	31
Scrum 24	Create login page UI	Feb 18	3	3
Scrum 23	Create registration page UI	Feb 18	2	2
Scrum 17	Create database schema (users, images tables)	Mar 13	5	5
Scrum 18	Build user registration API endpoint	Mar 13	3	3
Scrum 19	Build user login API with JWT	Mar 18	3	3
Scrum 20	Build image upload API endpoint	Mar 13	3	3
Scrum 21	Implement image preprocessing (resize, normalize)	Mar 3	2	2
Scrum 25	Build drag-and-drop image upload component	Mar 17	3	3
Scrum 26	Create image gallery view with thumbnails	Mar 17	2	2
Scrum 27	Implement JWT authentication flow	Mar 18	3	3
Scrum 28	Add form validation and error handling	Mar 18	2	2
US_07	**As an individual user, I want to receive an OA severity diagnosis.**		21	21
Scrum 35	Download OAI dataset (500+ images)	Mar 17	3	3
Scrum 36	Prepare and clean dataset	Mar 18	3	3

SPRINT 1 STORIES COMPLETED AND NOT

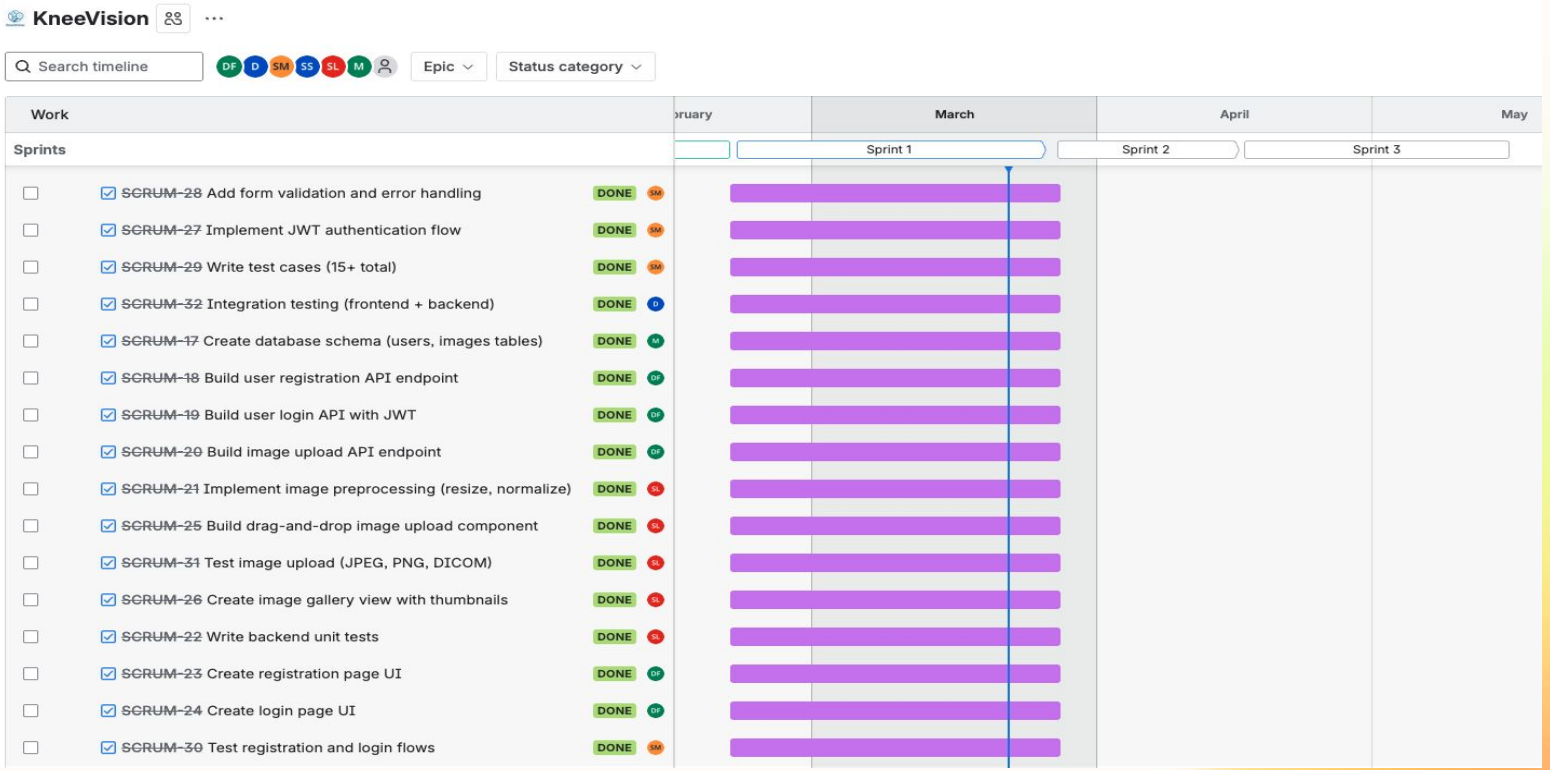
US_ID	User Stories	Completed Date	Story Points	Completed
Scrum 32	Integration testing (frontend + backend)	Mar 18	3	3
Scrum 31	Test image upload (JPEG, PNG, DICOM)	Mar 18	2	2
Scrum 30	Test registration and login flows	Mar 13	3	3
Scrum 29	Write test cases (15+ total)	Mar 13	3	3
Scrum 22	Write backend unit tests	Mar 18	2	2
Scrum 51	Start technical paper - Introduction	Mar 18	2	2

Sprint 1 Story Points: 52

SPRINT 0 PRE-PLANNING

SCRUM ID	Task	Assignee	Updated	Points
SCRUM-1	Define problem statement and project description	Daniel Fox	Feb 18	**5**
SCRUM-2	Create 3 user personas (Dr. Chen, Alex, Maya)	Manohar	Feb 18	**3**
SCRUM-3	Finalize technologies and algorithms	Deep	Feb 18	**1**
SCRUM-4	Finalize MVP overview	Daniel Fox	Feb 18	**2**
SCRUM-5	Finalize Team Working Agreement	Manohar	Feb 18	**1**
SCRUM-6	Finalize Team roles and responsibilities	Swatej Goud Mella	Feb 18	**2**
SCRUM-7	Finalize Project Schedule	Manohar	Feb 18	**2**
SCRUM-8	Sprint 0 Retrospective and recording	Daniel Fox	Feb 18	**1**
SCRUM-9	Finalize Sprint 0 Presentation	Deep	Feb 18	**3**
SCRUM-10	Sprint 0 Presentation and recording	Daniel Fox	Feb 19	**2**
SCRUM-11	Research knee OA datasets (OAI, MOST)	Sarvesh Shah	Feb 18	**5**
SCRUM-12	Create Jira Project and invite all teammates	Daniel Fox	Feb 18	**2**
SCRUM-13	Complete GitHub wiki page	Siming Li	Feb 19	**1**
SCRUM-14	Create Sprint 1 backlog, planning, and recording	Daniel Fox	Feb 19	**2**
SCRUM-16	Submit Sprint 0 deliverables and update wiki page	Daniel Fox	Feb 19	**2**

Project Schedule - Sprint 1



Product Backlog - Completed in Sprint 1

~~SCRUM-28~~	Add form validation and error handling	USER MANAGEMENT ...
~~SCRUM-27~~	Implement JWT authentication flow	USER MANAGEMENT ...
~~SCRUM-29~~	Write test cases (15+ total)	USER MANAGEMENT ...
~~SCRUM-32~~	Integration testing (frontend + backend)	USER MANAGEMENT ...
~~SCRUM-17~~	Create database schema (users, images tables)	USER MANAGEMENT ...
~~SCRUM-18~~	Build user registration API endpoint	USER MANAGEMENT ...
~~SCRUM-19~~	Build user login API with JWT	USER MANAGEMENT ...
~~SCRUM-20~~	Build image upload API endpoint	USER MANAGEMENT ...
~~SCRUM-21~~	Implement image preprocessing (resize, normalize)	USER MANAGEMENT ...
~~SCRUM-25~~	Build drag-and-drop image upload component	USER MANAGEMENT ...
~~SCRUM-31~~	Test image upload (JPEG, PNG, DICOM)	USER MANAGEMENT ...
~~SCRUM-26~~	Create image gallery view with thumbnails	USER MANAGEMENT ...
~~SCRUM-22~~	Write backend unit tests	USER MANAGEMENT ...
~~SCRUM-23~~	Create registration page UI	USER MANAGEMENT ...
~~SCRUM-24~~	Create login page UI	USER MANAGEMENT ...
~~SCRUM-35~~	Download OAI dataset (500+ images)	AI MODEL INTEGRAT...
~~SCRUM-36~~	Prepare and clean dataset	AI MODEL INTEGRAT...
~~SCRUM-30~~	Test registration and login flows	USER MANAGEMENT ...
~~SCRUM-51~~	Start technical paper - Introduction	AI MODEL INTEGRAT...

Product Backlog - Outstanding

SCRUM-33	Deploy to development environment	USER MANAGEMENT ...
SCRUM-34	Update API documentation (Swagger)	AI MODEL INTEGRAT...
SCRUM-37	Train ResNet50 model on Google Colab	AI MODEL INTEGRAT...
SCRUM-38	Evaluate model (target 75%+ accuracy)	AI MODEL INTEGRAT...
SCRUM-39	Save trained model (.h5 format)	AI MODEL INTEGRAT...
SCRUM-40	Create predictions table in database	AI MODEL INTEGRAT...
SCRUM-41	Build model serving endpoint	AI MODEL INTEGRAT...
SCRUM-42	Build /predict API endpoint	AI MODEL INTEGRAT...
SCRUM-43	Test prediction API (< 7 sec response)	AI MODEL INTEGRAT...
SCRUM-44	Store prediction results in database	AI MODEL INTEGRAT...
SCRUM-45	Build results display page	AI MODEL INTEGRAT...
SCRUM-46	Build results display page 1	AI MODEL INTEGRAT...
SCRUM-47	Show confidence score with progress bar	AI MODEL INTEGRAT...
SCRUM-48	Create KL grade explanation modal	AI MODEL INTEGRAT...
SCRUM-49	Add probability distribution visualization	AI MODEL INTEGRAT...
SCRUM-50	Integration testing (upload → predict → display)	AI MODEL INTEGRAT...
SCRUM-52	Technical paper - Methodology section	AI MODEL INTEGRAT...
SCRUM-53	Technical paper - Implementation section	AI MODEL INTEGRAT...

Product Backlog - Outstanding

☑ SCRUM-54	Create system architecture diagram	AI MODEL INTEGRAT...
☑ SCRUM-55	Document model training process	MVP COMPLETION & ...
☑ SCRUM-56	Build analysis history page	MVP COMPLETION & ...
☑ SCRUM-57	Add filter and sort functionality	MVP COMPLETION & ...
☑ SCRUM-58	Implement CSV export feature	MVP COMPLETION & ...
☑ SCRUM-59	Add educational tooltips throughout app	MVP COMPLETION & ...
☑ SCRUM-60	Make UI responsive (mobile/tablet/desktop)	MVP COMPLETION & ...
☑ SCRUM-61	Add error logging (backend + frontend)	MVP COMPLETION & ...
☑ SCRUM-62	Conduct UAT with 3-5 external testers	MVP COMPLETION & ...
☑ SCRUM-63	Cross-browser testing (Chrome, Firefox, Safari)	MVP COMPLETION & ...
☑ SCRUM-64	Performance testing (50 concurrent users)	MVP COMPLETION & ...
☑ SCRUM-65	Security audit (OWASP top 10)	MVP COMPLETION & ...
☑ SCRUM-66	Fix all critical and high-priority bugs	MVP COMPLETION & ...
☑ SCRUM-67	Deploy frontend to Vercel/Netlify	MVP COMPLETION & ...
☑ SCRUM-68	Deploy backend to Render/Railway	MVP COMPLETION & ...
☑ SCRUM-69	Deploy ML model endpoint	MVP COMPLETION & ...
☑ SCRUM-70	Set up monitoring and error tracking	MVP COMPLETION & ...
☑ SCRUM-71	Final end-to-end testing in production	MVP COMPLETION & ...

Product Backlog - Outstanding

SCRUM-72	Create privacy policy page	MVP COMPLETION & ...
SCRUM-73	Finalize technical paper (incorporate feedback)	MVP COMPLETION & ...
SCRUM-74	Write deployment manual (step-by-step)	MVP COMPLETION & ...
SCRUM-75	Write user manual with screenshots	MVP COMPLETION & ...
SCRUM-76	Record final MVP demo video (5-10 min)	MVP COMPLETION & ...
SCRUM-77	Prepare final presentation slides	MVP COMPLETION & ...

Test Cases

Test Summary

Module	Total	Passed	Failed	Coverage
Backend — imageProcessor.js	11	11	0	~95%
Backend — db.js	6	6	0	~90%
Backend — API routes (index.js)	17	17	0	~88%
Frontend — LoginForm	10	10	0	~92%
Frontend — RegisterForm	10	10	0	~91%
Frontend — ImageGallery	8	8	0	~90%
Frontend — ImageUploader	8	8	0	~85%
Frontend — Pages	16	16	0	~88%
TOTAL	86	86	0	~89%

Stack==> Node.js | Express | MongoDB | React | TypeScript | Jest | Vitest

Test Cases - Backend

Test Case	Test Name	Test Steps	Description / Expected Result	Status
1	Returns a Buffer	1. Create a 100×100 RGB PNG buffer using sharp. 2. Call preprocessImage(buffer). 3. Inspect the returned value.	preprocessImage() output is a Buffer instance	✓ PASS
2	Outputs PNG by default	1. Call preprocessImage(buffer) with no format option. 2. Inspect output metadata using sharp(result).metadata().	Default format produces image/png metadata	✓ PASS
3	Converts image to grayscale	1. Create a colour PNG buffer. 2. Call preprocessImage(buffer). 3. Read output metadata with sharp.	Output has 1 channel (grayscale)	✓ PASS
4	No upscale for small images	1. Call preprocessImage(buffer, { width: 1024, height: 1024 }). 2. Read output metadata.	50x50 image stays ≤50px when target is 1024x1024	✓ PASS
5	Resizes large image to fit target	1. Call preprocessImage(buffer, { width: 512, height: 512 }). 2. Read output metadata.	2000x2000 image resized to ≤512x512 when target is 512	✓ PASS
6	Preserves aspect ratio	1. Call preprocessImage(buffer, { width: 200, height: 200 }). 2. Compute width/height ratio of output.	200x100 (2:1) input produces ~2:1 ratio output	✓ PASS
7	Outputs JPEG when format=jpeg	1. Call preprocessImage(buffer, { format: 'jpeg' }). 2. Read output metadata.	format:'jpeg' option produces jpeg metadata	✓ PASS
8	Outputs WebP when format=webp	1. Call preprocessImage(buffer, { format: 'webp' }). 2. Read output metadata.	format:'webp' option produces webp metadata	✓ PASS
9	Normalize option does not throw	1. Call preprocessImage(buffer, { normalize: true }). 2. Await the result.	normalize:true runs without error	✓ PASS
10	Throws on invalid buffer	1. Create Buffer.from('not-an-image'). 2. Call preprocessImage(badBuffer). 3. Await and expect rejection.	Non-image buffer rejects with 'Image preprocessing failed'	✓ PASS
11	Throws on null input	1. Call preprocessImage(null). 2. Await and expect rejection.	null input causes rejection	✓ PASS
12	Throws when MONGODB_URI missing	1. Delete process.env.MONGODB_URI. 2. Call connectToMongo(). 3. Await the promise.	connectToMongo() rejects with 'MONGODB_URI is not set'	✓ PASS
13	Connects and returns db instance	1. Set process.env.MONGODB_URI to a valid URI. 2. Call connectToMongo(). 3. Inspect return value.	Valid URI resolves to a db object	✓ PASS
14	Creates unique index on users.email	1. Call connectToMongo(). 2. Check that db.collection('users').createIndex was called.	createIndex called with {email:1}, {unique:true}	✓ PASS
15	Caches db on repeated calls	1. Call connectToMongo() twice. 2. Check client.connect call count.	client.connect called exactly once after two connectToMongo() calls	✓ PASS

Test Cases - Backend

Test Case	Test Name	Test Steps	Description / Expected Result	Status
16	getDb throws before connect	1. Import getDb from fresh module. 2. Call getDb() immediately.	getDb() throws 'MongoDB not connected' before connectToMongo()	✓ PASS
17	getDb returns db after connect	1. Call connectToMongo(). 2. Call getDb(). 3. Inspect return value.	getDb() returns db object after successful connect	✓ PASS
18	GET /health returns 200	1. Send GET request to /health.	Response body is {status:'ok'}	✓ PASS
19	Register — valid input returns 201	1. POST /api/v1/auth/register with { email: 'test@example.com', password: 'password123' }. 2. Inspect response.	Returns id, email, createdAt on valid body	✓ PASS
20	Register — missing email → 400	1. POST /api/v1/auth/register with { password: 'password123' } only.	Returns 400 with 'required' message	✓ PASS
21	Register — missing password → 400	1. POST /api/v1/auth/register with { email: 'test@example.com' } only.	Returns 400	✓ PASS
22	Register — invalid email → 400	1. POST /api/v1/auth/register with { email: 'not-an-email', password: 'password123' }.	Returns 400 with 'Invalid email format'	✓ PASS
23	Register — short password → 400	1. POST /api/v1/auth/register with { email: 'test@example.com', password: 'short' }.	Returns 400 with '8 characters' message	✓ PASS
24	Register — duplicate user → 409	1. POST /api/v1/auth/register with existing email.	Returns 409 with 'User already exists'	✓ PASS
25	Register — email normalized	1. POST with { email: 'USER@EXAMPLE.COM', password: 'password123' }. 2. Check response body email field.	USER@EXAMPLE.COM stored as user@example.com	✓ PASS
26	Login — valid credentials → 200	1. POST /api/v1/auth/login with correct email and password.	Returns JWT token and user object	✓ PASS
27	Login — empty body → 400	1. POST /api/v1/auth/login with empty body {}.	Returns 400	✓ PASS
28	Login — unknown user → 401	1. POST /api/v1/auth/login with unregistered email.	Returns 401 with 'Invalid email or password'	✓ PASS
29	Login — wrong password → 401	1. POST /api/v1/auth/login with correct email but wrong password.	Returns 401	✓ PASS
30	Presign — valid input → 200	1. POST /api/v1/uploads/presign with { filename: 'xray.png', contentType: 'image/png' }.	Returns uploadUrl, fileUrl, method:PUT	✓ PASS

Test Cases - Backend

Test Case	Test Name	Test Steps	Description / Expected Result	Status
31	Presign — missing filename → 400	1. POST /api/v1/uploads/presign with { contentType: 'image/png' } only.	Returns 400	✓ PASS
32	Presign — missing contentType → 400	1. POST /api/v1/uploads/presign with { filename: 'xray.png' } only.	Returns 400	✓ PASS
33	Images — valid input → 201	1. POST /api/v1/images with { fileUrl: '/uploads/test.png', originalName: 'xray.png', contentType: 'image/png' }.	Returns id, fileUrl, originalName, contentType	✓ PASS
34	Images — missing fileUrl → 400	1. POST /api/v1/images with { originalName: 'xray.png' } only.	Returns 400 with 'fileUrl is required'	✓ PASS

Test Cases - Frontend

Test Case	Test Name	Test Steps	Description / Expected Result	Status
1	Renders all form fields	1. Render <LoginForm onLoginSuccess={fn} />. 2. Query for email input, password input, Login button.	Email, password inputs and Login button present	✓ PASS
2	Empty email shows error	1. Click the Login button without filling any field.	Submitting with no email shows 'Email is required'	✓ PASS
3	Empty password shows error	1. Type a valid email. 2. Click Login without filling password.	Submitting with no password shows 'Password is required'	✓ PASS
4	No fetch on validation failure	1. Click Login with empty fields.	fetch() not called when form is invalid	✓ PASS
5	Fetch called with correct payload	1. Type 'User@Example.COM' in email field. 2. Type 'password123' in password field. 3. Click Login.	Email lowercased, sent to /api/v1/auth/login	✓ PASS
6	Token stored in localStorage	1. Submit valid credentials. 2. Check localStorage after submission.	localStorage.getItem('token') equals returned JWT	✓ PASS
7	onLoginSuccess called on success	1. Submit valid credentials. 2. Wait for async resolution.	Callback fires once after successful login	✓ PASS
8	Server error shown	1. Submit credentials. 2. Wait for response.	'Invalid email or password' shown on 401 response	✓ PASS
9	Network error shown	1. Submit valid credentials.	'Network Error' shown when fetch throws	✓ PASS
10	Button disabled while loading	1. Submit valid credentials. 2. Immediately inspect button state.	Submit button is disabled while fetch is pending	✓ PASS
11	Renders all form fields	1. Render <RegisterForm onRegistrationSuccess={fn} />. 2. Query for email, password, confirm password, Register button.	Email, password, confirm password, Register button present	✓ PASS
12	Empty email shows error	1. Click Register without filling any field.	'Email is required' shown on empty submit	✓ PASS
13	Short password shows error	1. Type email. 2. Type password shorter than 8 characters. 3. Click Register.	'at least 8 characters' shown for short password	✓ PASS
14	Password mismatch shows error	1. Fill email and password. 2. Type different value in confirm password. 3. Click Register.	'Passwords do not match' shown when fields differ	✓ PASS
15	No fetch on validation failure	1. Click Register with empty form.	fetch() not called when form is invalid	✓ PASS

Test Cases - Frontend

Test Case	Test Name	Test Steps	Description / Expected Result	Status
16	Fetch called with correct payload	1. Fill 'User@Example.COM', 'password123', 'password123'. 2. Click Register.	Email lowercased, sent to /api/v1/auth/register	✓ PASS
17	Success message and redirect	1. Submit valid form. 2. Advance timers by 800ms.	'Registration successful' shown; onRegistrationSuccess fires after 800ms	✓ PASS
18	Form cleared on success	1. Submit valid form. 2. Wait for success state.	All fields empty after successful registration	✓ PASS
19	Server error shown	1. Submit valid form.	'User already exists' shown on 409 response	✓ PASS
20	Button disabled while submitting	1. Submit valid form. 2. Inspect button state.	Register button disabled while fetch pending	✓ PASS
21	Empty state message shown	1. Render <ImageGallery images={[]} />.	'No images to display' shown for empty array	✓ PASS
22	Empty state for undefined prop	1. Render <ImageGallery images={undefined} />.	'No images to display' shown when images is undefined	✓ PASS
23	Renders img per image	1. Render <ImageGallery images={[img1, img2]} />. 2. Query all <img> elements.	One <img> element rendered per image with url	✓ PASS
24	title used as alt text	1. Render gallery with image { url: '...', title: 'Knee X-ray' }.	img alt equals image.title	✓ PASS
25	Falls back to originalName	1. Render gallery with image { url: '...', originalName: 'scan.png' }.	img alt equals originalName when title absent	✓ PASS
26	Falls back to fileUrl	1. Render gallery with image { fileUrl: '/uploads/scan.png', title: 'Scan' }.	img src equals fileUrl when url absent	✓ PASS
27	No preview placeholder	1. Render gallery with image { id: '1', title: 'Mystery' }.	'No preview' shown when url and fileUrl both absent	✓ PASS
28	Label text rendered below image	1. Render gallery with image { url: '...', title: 'Knee Scan' }.	Title/name text visible below the image	✓ PASS
29	Renders dropzone text	1. Render <ImageUploader />.	'Drag & drop or click' text visible	✓ PASS
30	Shows drag-active text	1. Fire dragOver event on the dropzone element.	'Drop files here' shown on dragenter	✓ PASS

Test Cases - Frontend

Test Case	Test Name	Test Steps	Description / Expected Result	Status
31	No upload button initially	1. Render <ImageUploader /> without dropping files.	Upload button absent before files are dropped	✓ PASS
32	Upload button after drop	1. Fire drop event with a PNG file. 2. Query for Upload button.	Upload button appears after files are dropped	✓ PASS
33	DICOM placeholder shown	1. Fire drop event with a .dcm file. 2. Query for DICOM label.	'DICOM' placeholder shown for .dcm files	✓ PASS
34	Full upload flow succeeds	1. Drop a PNG file. 2. Click Upload. 3. Wait for all three fetch calls.	presign → PUT → register called in order; onUploadSuccess fires	✓ PASS
35	Alert on presign failure	1. Drop a file. 2. Click Upload.	window.alert called when presign returns non-ok	✓ PASS
36	Button disabled while uploading	1. Drop a file. 2. Click Upload. 3. Inspect button state.	Upload button disabled while fetch is pending	✓ PASS
37	LoginPage — brand heading	1. Render <LoginPage onLoginSuccess={fn} onSwitchToRegister={fn} />.	'KneeVision' h1 rendered	✓ PASS
38	LoginPage — subtitle	1. Render LoginPage.	'Sign in to your account' text present	✓ PASS
39	LoginPage — form rendered	1. Render LoginPage.	Email and password inputs present inside page	✓ PASS
40	LoginPage — switch to register	1. Click the Register button.	onSwitchToRegister called when Register button clicked	✓ PASS
41	RegisterPage — brand heading	1. Render <RegisterPage onRegistrationSuccess={fn} onSwitchToLogin={fn} />.	'KneeVision' h1 rendered	✓ PASS
42	RegisterPage — subtitle	1. Render RegisterPage.	'Create your account' text present	✓ PASS
43	RegisterPage — form rendered	1. Render RegisterPage.	Email and confirm password inputs present inside page	✓ PASS
44	RegisterPage — switch to login	1. Click the Login button.	onSwitchToLogin called when Login button clicked	✓ PASS
45	UploadPage — brand heading	1. Render <UploadPage onLogout={fn} />.	'KneeVision' h1 rendered	✓ PASS

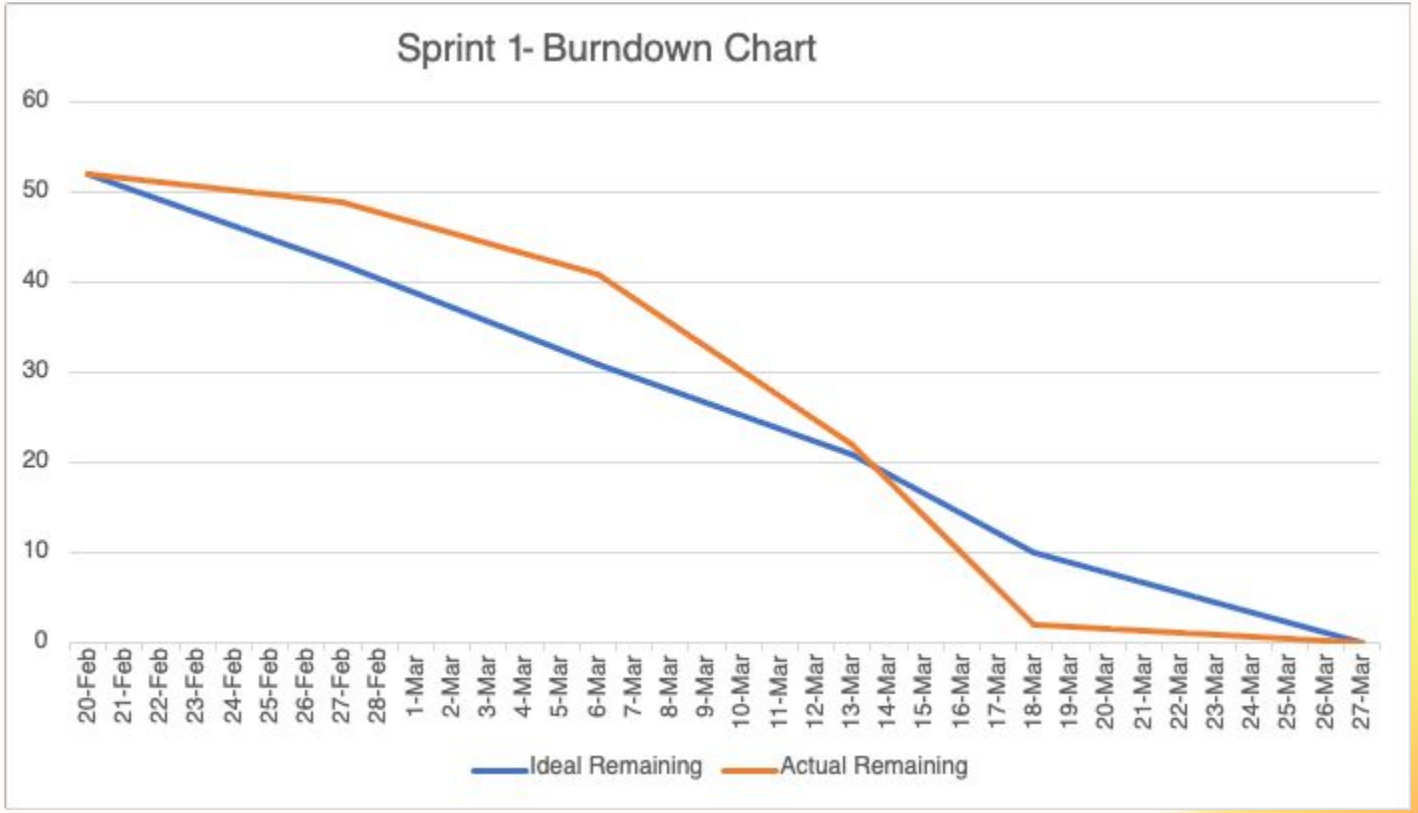
Test Cases - Frontend

Test Case	Test Name	Test Steps	Description / Expected Result	Status
46	UploadPage — subtitle	1. Render UploadPage.	'Upload and review knee X-ray' text present	✓ PASS
47	UploadPage — logout button	1. Render UploadPage.	Logout button visible	✓ PASS
48	UploadPage — logout callback	1. Click the Logout button.	onLogout called when Logout clicked	✓ PASS
49	UploadPage — gallery heading	1. Render UploadPage.	'Gallery' heading present	✓ PASS
50	UploadPage — empty state	1. Render UploadPage with no images.	'No images to display yet' shown initially	✓ PASS
51	UploadPage — add mock image	1. Click 'Add Mock Image'. 2. Query for img element.	Mock image appears in gallery after clicking 'Add Mock Image'	✓ PASS
52	UploadPage — empty state hidden	1. Click 'Add Mock Image'. 2. Query for empty state text.	Empty state message gone after image added	✓ PASS

Metrics - Team Velocity

52 story points
completed

Metrics - Burndown Chart



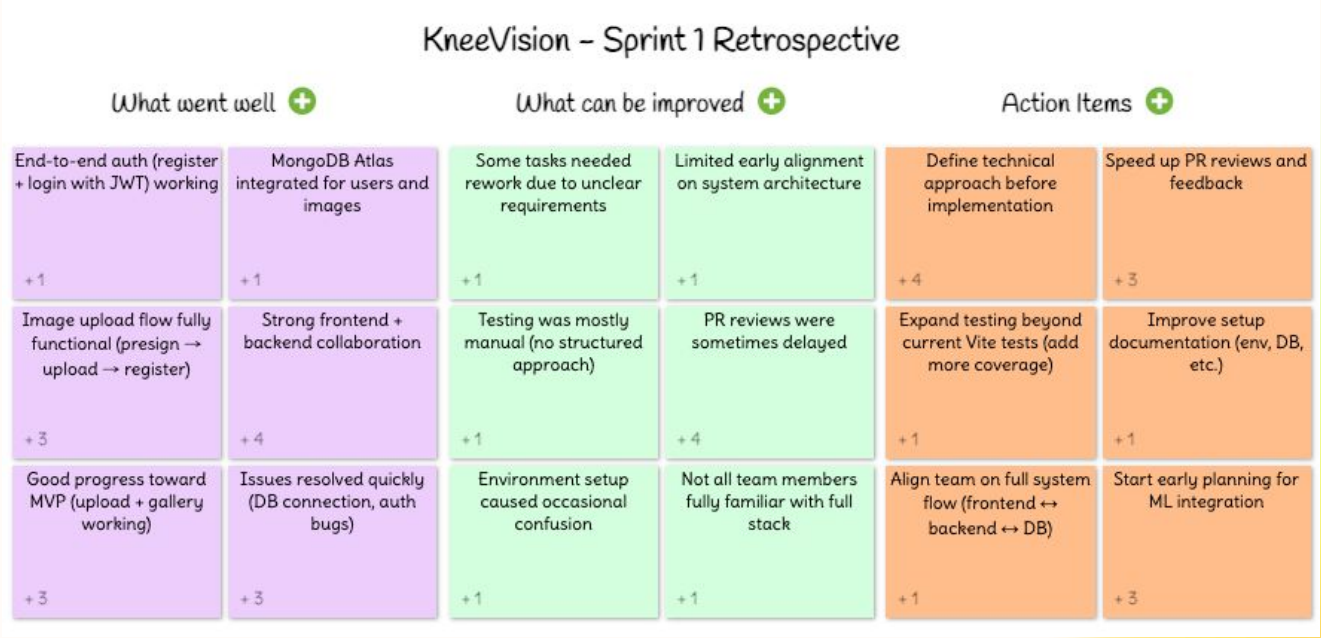
Metrics - Completed/Committed Ratio

52 / 52

100%

Sprint 1 Story Points: 52

Sprint 1 Retrospective



Sprint 1 Retrospective: Important Points

What Went Well

- Strong frontend + backend collaboration
- Image upload flow fully functional
- Good progress toward MVP
- Issues resolved quickly

Improvements

- PR reviews were sometimes delayed

Action Items

- Define technical approach before implementation
- Start early planning for ML integration

Project Roadmap

Sprint Schedule & Key Deliverable

0

DEADLINE: FEB 19

Foundation

- Project Charter & MVP
- GitHub & Tech Stack Setup
- Backlog Creation

DELIVERABLES:
Presentation + Wiki + Backlog

1

DEADLINE: MAR 26

Core Features

- User Auth
- Image Upload
- Image Gallery

DELIVERABLES:
Working Demo + Wiki + Tests

2

DEADLINE: APR 16

AI Engine

- Model Training
- Prediction API
- KL Grade Results

DELIVERABLES:
AI Demo + Draft Paper + Wiki

3

DEADLINE: MAY 14

Deployment

- History & CSV Export
- Security & UAT Audit
- Production Launch

DELIVERABLES:
Final MVP + Full Documentation

SPRINT 2 - USER STORIES

US_ID	User Stories	Acceptance Criteria	Feature	Story Points
US_09	As an individual user, I want to receive a brief description of the diagnosed condition so that I can understand the issue I am facing.	After receiving the diagnosis, the user is provided with a brief description of the diagnosed condition. The description is written in clear, understandable language. The description includes common symptoms and causes.	AI Model Integration	2
US_02	As a research coordinator, I want to upload multiple X-ray images at once so that I can analyze datasets efficiently.	The patient can upload multiple images (up to 3) to one case. The system accepts multiple images in specified formats (JPEG, JPG, PNG). All uploaded images are associated with the same patient case. The patient receives a confirmation message after uploading each image. The system displays a preview of all uploaded images for review before analysis.	Image Upload	8
US_08	As an individual user, I want to receive the most accurate possible diagnosis so that I can trust the assessment provided.	The system continuously improves the model and algorithms for better diagnostic results. The diagnosis accuracy is optimised based on model refinements and external validation. The confidence level of the diagnosis should be high (e.g., 95%).	AI Model Integration	11
US_10	As an individual user, I want to receive treatment recommendations so that I can take action towards treating my knee condition.	After receiving the diagnosis, the user is provided with treatment recommendations. The description is written in clear, understandable language. The description includes common symptoms and causes of the condition.	AI Model Integration	3

Project - APIs - Login API

```
app.post("/api/v1/auth/login", async (req, res) => {
  const { email, password } = req.body || {};
  const normalizedEmail = String(email || "").trim().toLowerCase();

  const db = getDb();
  const users = db.collection("users");
  const user = await users.findOne({ email: normalizedEmail });

  if (!user) {
    return res.status(401).json({ message: "Invalid email or password." });
  }

  const passwordMatches = await bcrypt.compare(password, user.passwordHash);
  if (!passwordMatches) {
    return res.status(401).json({ message: "Invalid email or password." });
  }

  const token = jwt.sign(
    { userId: user.userId || user._id.toString(), email: user.email },
    process.env.JWT_SECRET || "dev-secret-change-me",
    { expiresIn: "1h" }
  );

  return res.status(200).json({ token, user: { userId: user.userId, email: user.email } });
});
```

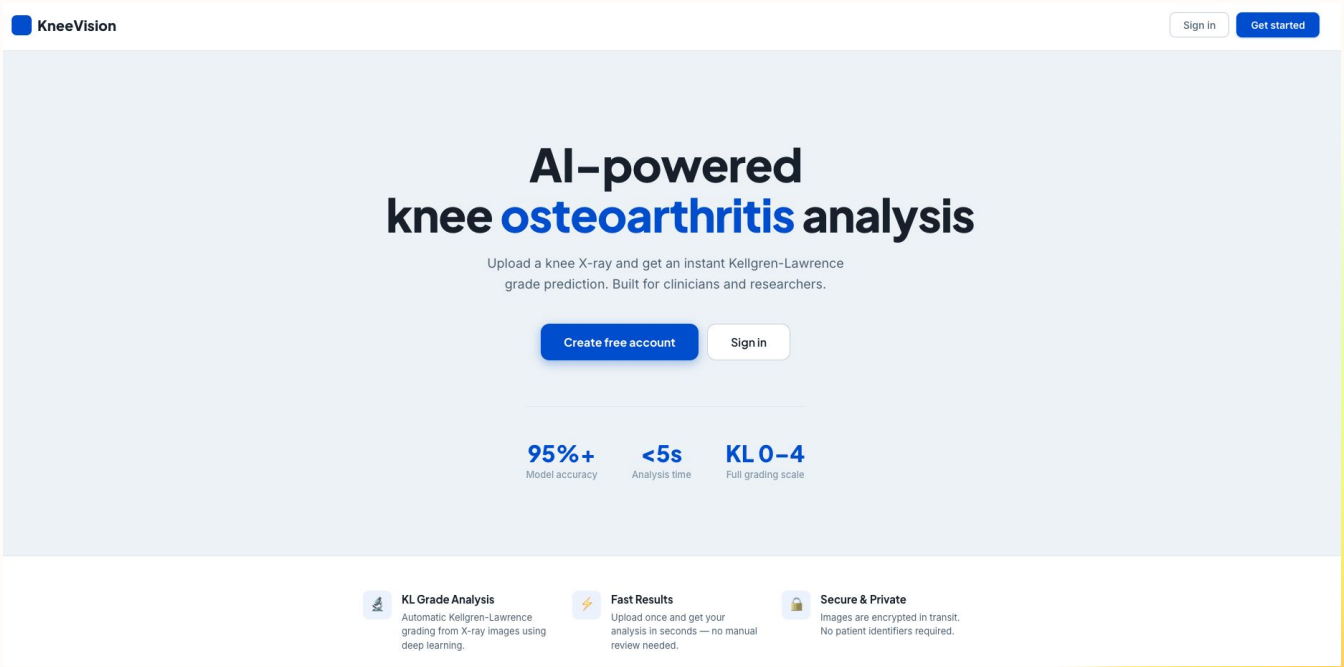
Project - APIs - Upload Image API

```
app.post("/api/v1/images", requireAuth, async (req, res) => {
  const { fileUrl, originalName, contentType } = req.body || {};

  const imageDoc = {
    fileUrl,
    originalName: originalName || null,
    contentType: contentType || null,
    uploadedBy: {
      userId: req.user.userId,
      email: req.user.email,
    },
    createdAt: new Date().toISOString(),
    updatedAt: new Date().toISOString(),
  };

  const db = getDb();
  const images = db.collection("images");
  const result = await images.insertOne(imageDoc);

  return res.status(201).json({
    id: result.insertedId.toString(),
    fileUrl: imageDoc.fileUrl,
    uploadedBy: imageDoc.uploadedBy,
  });
});
```

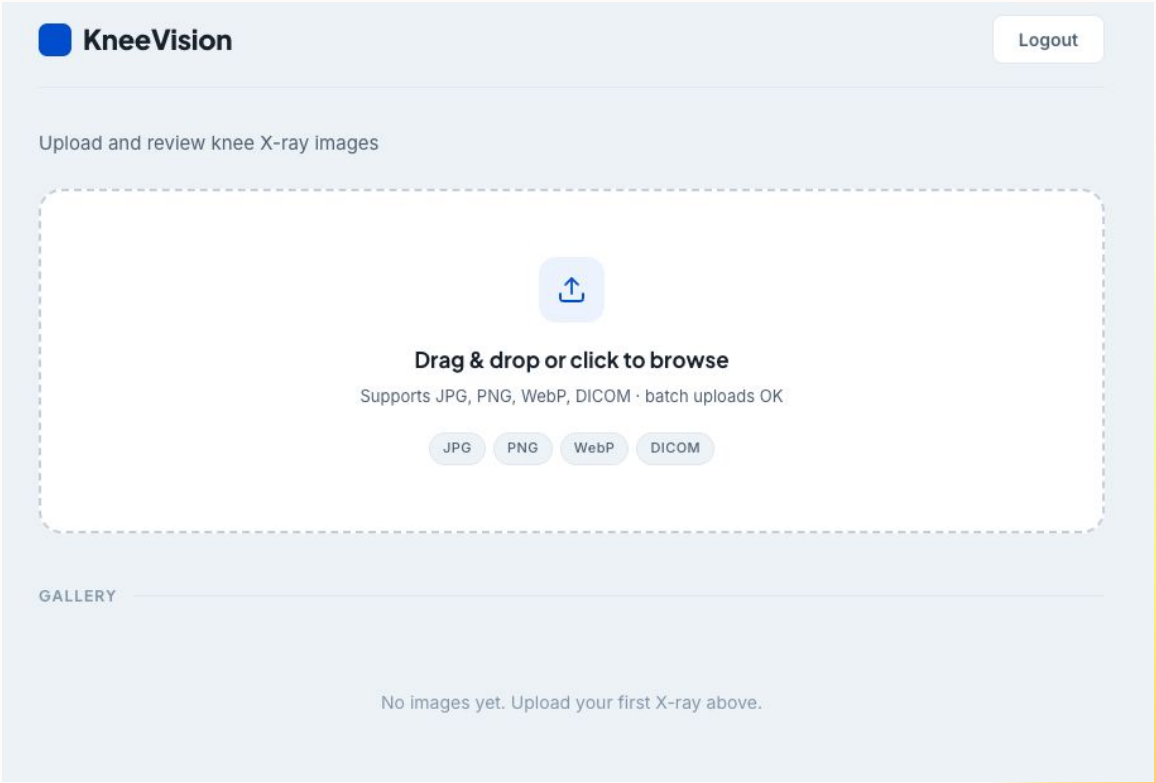
Project - APIs - Get Images API

```
app.get("/api/v1/images", requireAuth, async (req, res) => {
  const db = getDb();
  const images = db.collection("images");

  const docs = await images
    .find({ "uploadedBy.userId": req.user.userId })
    .sort({ createdAt: -1 })
    .toArray();

  return res.status(200).json(
    docs.map((doc) => ({
      id: doc._id.toString(),
      fileUrl: doc.fileUrl,
      originalName: doc.originalName,
      uploadedBy: doc.uploadedBy,
      createdAt: doc.createdAt,
    }))
  );
});
```

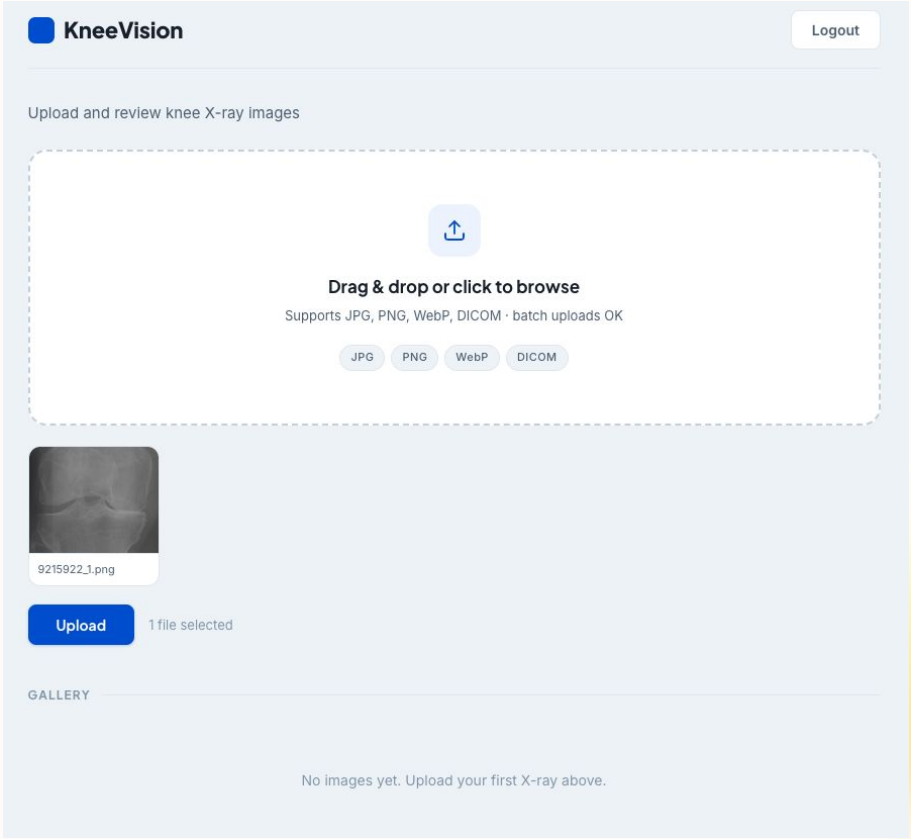
Project Demo - Application Screenshots



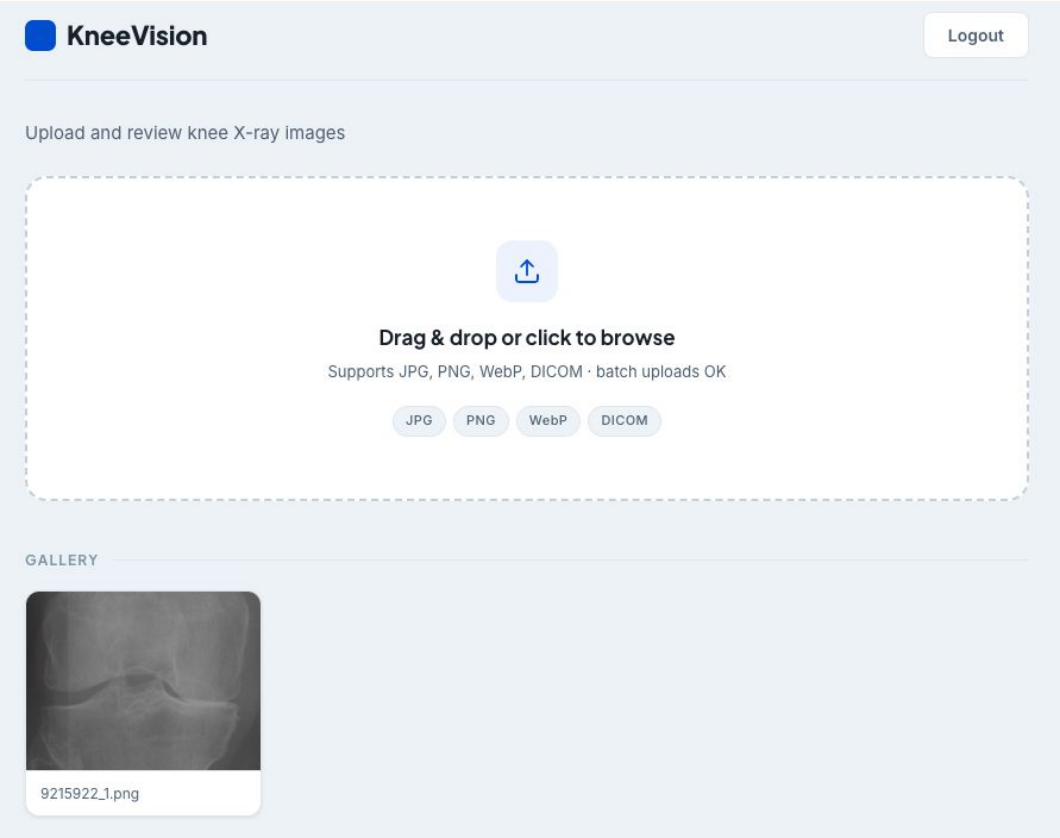
Project Demo - Application Screenshots



Project Demo - Application Screenshots



Project Demo - Application Screenshots



Wiki Page Link

https://github.com/htmw/STARLABS/wiki#kneevision

Live Application Demo

AI Disclosure

- Assisted in generating initial user stories and backlog items
- Helped draft test cases and testing scenarios
- Supported development of specific backend API routes (authentication, image upload, and database integration) and frontend components (forms, state handling, and API calls)
- Provided guidance for debugging and resolving errors
- Assisted with structuring documentation and project artifacts

THANK YOU

Team S.T.A.R Labs